

6.4.1 Schrödinger's One-dimensional Time Dependent Wave Equation

For deriving time, dependent Schrödinger's one dimensional wave equation let us eliminate the total energy E from time, independent wave equation. The wave function is given by

$$\psi(x, t) = Ae^{i(kx - \omega t)} \quad (23)$$

Differentiating the above equation w.r.t 't', we get

$$\begin{aligned} \frac{d\psi}{dt} &= -i\omega Ae^{i(kx - \omega t)} \\ &= -i(2\pi\nu) Ae^{i(kx - \omega t)} \\ &= -2\pi i\nu \psi \end{aligned} \quad (\because \omega = 2\pi\nu)$$

Since

$$\begin{aligned} E &= h\nu, \nu = \frac{E}{h} \\ \frac{d\psi}{dt} &= -\frac{2\pi i E}{h} \psi \end{aligned}$$

Since

$$\begin{aligned} \hbar &= \frac{h}{2\pi}, h = 2\pi\hbar \\ \frac{d\psi}{dt} &= -\frac{2\pi i E}{2\pi\hbar} \psi \\ \frac{d\psi}{dt} &= \frac{-iE}{\hbar} \psi = \frac{E\psi}{i\hbar} \\ E\psi &= i\hbar \frac{d\psi}{dt} \end{aligned} \quad (24)$$

Substituting the value of $E\psi$ in the equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

we get
$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left[i\hbar \frac{d\psi}{dt} - V\psi \right] = 0$$

$$\frac{d^2\psi}{dx^2} = \frac{-2m}{\hbar^2} \left[i\hbar \frac{d\psi}{dt} - V\psi \right]$$

$$\frac{-\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = i\hbar \frac{d\psi}{dt} \quad (25)$$

The above equation represents time, dependent Schrödinger's one-dimensional wave equation. For three-dimensional motion, the above equation becomes

$$\frac{-\hbar^2}{2m} \nabla^2 \psi + V\psi = i\hbar \frac{d\psi}{dt}$$

where

$$\nabla^2 = \frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \quad (26)$$

Physical Significance of Wave Function ψ